Automatic Fusion of Satellite Imagery and AIS data for Vessel Detection

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***Abstract*—Being able to fuse information coming from different sources, such as AIS and satellite images is of major importance for maritime domain awareness, for example, to locate and identify vessels that may have purposely turned off their AIS transponder, preventing illegal activities. This paper presents a fully-automatic method for fusing AIS data and SAR satellite images for vessel detection. The proposed framework is based on the automatic annotation of satellite images by correlating them with AIS data producing train and test datasets which are provided as input to a convolutional neural network (CNN). The CNN was trained to detect the presence of ships in sectors of the image. Our automatic process allows the neural network to learn on a large amount of data, without the need for hand-labelled datasets. Our neural network, trained on our automatically-generated test set of images, achieved an accuracy of 88% at ship detection, and an area under ROC curve of 94,6%. Our estimate of real world accuracy is about 86-90%.**

I. Introduction

Maritime domain awareness is becoming an increasingly important endeavor for governments of coastal countries around the world. The establishment of the Automatic Identification System (AIS) by the International Maritime Organization as a standard was a significant milestone in the maritime domain, as it became mandatory for a large number of vessels which could then be tracked down. Despite the significant contribution of the AIS standard to the maritime domain, using it as the only source of information about the locations of vessels is often not enough, as (i) only vessels weighted above 300 tonnes are obligated to have an AIS transponder, (ii) there are not enough AIS receivers around the globe, and (iii) vessels involved in illegal activities frequently turn off their AIS transponder intentionally. Recent reports suggest that almost 90% of illegal vessel activity remains undetected [1]. Since increasing maritime awareness is of paramount importance for stakeholders, such as companies and governments, a lot of research and industrial efforts are focused on the development of techniques and products for the integration of data coming from multiple sources.

At the same time, the amount and the availability of satellite data is increased in the recent years. Copernicus[1] is a large European programme in the context of which last generation satellites, called Sentinels, went into orbit in order to monitor the Earth and make high resolution data available to users and the general public. The Copernicus constellation of satellites collects approximately 10 terabytes (TB) of data on a daily basis. Using high resolution imagery could be very useful when used combined with other sources of data, such as AIS data, in order to improve maritime situational awareness.

In the present paper we describe our fully automatic framework that is based on neural networks for performing vessel detection on satellite images using AIS data. The contributions of the proposed approach are the following:

- It proposes a fully automatic workflow of fusing AIS data with satellite data in general (not only SAR data)
- It employs deep learning techniques in order to make predictions about the existence or not of vessels in parts of satellite imagery
- It performs experimental evaluation using real AIS and satellite data. Our proposed framework is orthogonal to approaches proposed in literature in that it can be used in a complimentary manner and be easily embedded in their workflows.

The paper is organised as follows. Section II presents related work in fusing AIS data with satellite imagery. Section III provides explanations on some basic background terminology and vocabulary that is used throughout the paper, for the convenience of the reader. Section IV describes our approach and Section V presents the results of our experimental evaluation. Section VI details the structure of the convolutional neural network that was used, and Section VII presents the results of the experimental evaluation together with some discussion of what the results mean. Last, Section VIII concludes the paper and expands on potential future extensions of this work.

II. Related work

Several different approaches have been attempted at fusional strategies combining AIS data with other sources. A frequent choice of alternative data source for fusion is SAR imagery, which has been used in several different papers.

The authors of [2] propose a simplified scenario of AIS and SAR fusion, modeled as a binary hypothesis testing problem. Synthetic sensor data is used to prove the validity of the

[1]https://www.copernicus.eu/en

algorithmic approach. In the work described in [3], AIS signals are correlated with SAR images after an identification process of each ship through the AIS signals not based on the self-reported potentially-faulty MMSI ship identifier, but rather identification based on characteristics of the ship movement itself (track, etc.) so as to be more accurate. The correlation is then performed with SAR via a process of simulated annealing (SA). The work described in [4] introduces a model for optimising the SAR tracking strategy, as opposed to random SAR tracking, in order to increase the performance of multi-sensor maritime surveillance. In [5], a method is proposed in SAR cases of high wave interference, where detection is quite difficult due to the high noise of the background in relation to the ship itself in the SAR image. The method uses an adaptive threshold in conjunction with local contrast variance weighted information entropy. In [6], Brazilian researchers examined the Campos Basin in Brazil, through the fusion of SAR imagery from the TerraSAR satellite and AIS data to detect ships and offshore oil platforms. They utilized the Constant False Alarm Rate (CFAR) algorithm to perform the detection itself, which is based on the relative contrast of the ship with its surrounding background in the image, achieving accuracy of 82% on average.

In [7], the tool SUMO was used to detect ships from Sentinel-1 SAR imagery. The SUMO tool utilizes the CFAR algorithm for detection, and the coastline of India was used as a testing area. In [8], a better method for correlation of AIS and SAR positions is described, through a more effective method of interpolation of the ship positions from the AIS signals to the moment that the SAR image was taken. This method involves using the regional traffic to better predict the future or past position of the ship, relative to the moment of the SAR image. In the work described in [9] the authors compare C-Band SAR images for a specific area acquired by Sentinel-1 with X-Band images for the same area acquired by TerraSAR-X. The authors propose an algorithm for ship detection that is based on the likelihood functions of both canonical ship target and sea clutter. In order to evaluate the algorithm against C-Band and X-Band SAR images, the authors used AIS data received the same time as the acquisition of the image as ground truth. In [10] the authors describe a workflow for fusing AIS data with SAR images. More specifically, they first apply the CFAR algorithm on the SAR image and then, they extract information about the vessels. Next, they import the respective information from AIS data and they build separate vectors for each of these sources. For the evaluation, simulated AIS and SAR data were used. In work described in [11], the authors propose a methodology called distributed multi-hypothesis tracking (DMHT) for using data from heterogeneous sources, such as AIS tracks, coastal radar contacts, and SAR imagery for maritime surveillance. The evaluation of the proposed approach was based on simulated data, in the same way as the approaches discussed above.

III. Background

AIS (Automatic Identification System) is a system initially designed for collision avoidance between ships on the high seas. Each ship has on board a transponder, which transmits three types of data: static ship-related, route-related, and dynamic ship-related. Static ship-related includes information such as the MMSI (Maritime Mobile Service Identity) of the ship, its dimensions, and its type. Route-related includes voyage information such as the destination and port of departure of the ship, the type of cargo it holds, etc. Finally, dynamic ship-related includes the ship's GPS position, its speed, its heading, etc. Generally, the final category is the most relevant to detection research and maritime awareness.

AIS was made mandatory in 2004 by the International Maritime Organization for ships above a certain weight (300 tonnes). Since the AIS standard is an open and well documented standard and since anyone can collect the AIS signals that are sent out by ships, the AIS system has more recently found a use in maritime surveillance and awareness, through a network of receivers that collect the signals sent out by the ships and provide this vast wealth of data as a service users can access (e.g. MarineTraffic). Simultaneously, due to this open nature of AIS, they are frequently used for scientific and research purposes due to their easy availability. For instance, a recent paper [12] describes how AIS data can be used to detect the loitering phenomenon - a kind of illegal, unreported, and unregulated fishing, a major environmental and economic problem at a global scale.

There are two primary types of AIS receiver networks, terrestrial AIS and satellite AIS. Terrestrial AIS refers to signals that are received by terrestrial receivers, that is to say on the ground, on coastlines around the world. Since the AIS signal has an unfortunate range limit of 15-60 nautical miles, depending on altitude, this cannot cover the entirety of the earth in terms of AIS coverage. For this purpose, satellite AIS comes in, which refers to the AIS signals that are picked up by satellite receivers to fill in for the lack of terrestrial coverage [13].

Fusion of multiple sources in the domain of maritime awareness is extremely important. As mentioned previously, in order to get a more complete picture of the maritime environment, one cannot rely only on AIS data. Not only can ships disable their AIS transponders in the event that they wish to commit illegal activity, but the existence of dark spots (ships that are not carrying AIS transponders for various reasons) cannot be discounted. For this reason, governments and agencies around the world are seeking to augment the picture of the maritime environment that AIS gives using alternative sources of data, as for example, satellite imagery. Satellite imagery primarily allows for the detection of instances where ships have disabled their AIS purposefully to engage in illegal activity. As such, when the AIS transponder is switched off, the ship will still be visible in the satellite imagery, and be able to be identified.

As the image processing and classification part of the system presented here is handled using Convolutional Neural Networks, the remainder of this section discuses these algorithms and the tools that provide them. Convolutional Neural Networks are a class of deep learning neural networks that have attracted the attention of research and industry over the past few years, mainly due to their high accuracy when applied to analysing visual imagery. In [14] convolutional nets were

tested against the ImageNet dataset for object recognition, and achieved a significant reduction in the error rate for object recognition, leading to the rapid adoption of deep learning by the computer vision community. In [15] the authors investigated a semi-supervised learning representation by considering features extracted by a deep convolutional network for the purposes of completing generic tasks such as scene recognition, domain adaptation, and fine-grained recognition. The solution proposed in [14] is documented in [16] where a novel visualisation technique, one that gives insights on the functionality and the inherent flaws of intermediate layers of CNNs and the operation of the classifier, is introduced.

In [17] the authors proposed a network architecture based on dilated convolutions and dense connections among all feature maps, achieving in this way the reduction of the number of feature maps and trainable parameters without a corresponding fall in accuracy. The solution focuses on problems with dense outputs, in which the number of rows and columns of the output image is identical to those of the input image. The authors in [18] use the CNN implementation of [14] to extract features and locate objects in image regions. They also show that a supervised pre-training process can boost system performance when labeled training data is scarce. ResNet [19] is an very deep neural network with 152 layers proposed by Microsoft, that won the ILSVRC and COCO 2015 competition,s beating its main competitors at all the main tasks (classification, detection, localisation and segmentation). It was considered a revolution of depth when compared to GoogleNet [20], VGG [21], ZFNet [16] and AlexNet [14], systems that had 22, 19, 8 and 8 layers respectively. The authors showed that going deeper is not always the best practice, as they observed that accuracy gets saturated and then degrades rapidly when the network depth is increased too much. To avoid this issue, they explicitly let layers in their network fit a residual mapping, assuming that if the identity mapping was optimal it would be easier to push the residual to zero than to fit an identity mapping by a stack of nonlinear layers. However, all these CNN models have been trained with large datasets that have high variability in their labelled data. For instance, ImageNet has over 14 million images with 1000 labels for classification. Thus, although these models are capable to distinguish generic classes (e.g., animal, plant, person, ship). they are not domain-specific and will fail to identify sub-classes such as Containers, Passenger Ships and Tankers in case of ships. This is reasonable, as it is difficult to find large volumes of data that are specific to a domain. Some recent approaches that attempt to exploit the vast increase of Sentinel images and try to identify objects from SAR images [22][23] or aerial images of the visible spectrum [24] have emerged. Although the results are encouraging, those solutions are still simply identifying if an object is a ship or not, while in [25] the authors use TerraSAR-X high resolution images to perform a primitive classification of five maritime classes, namely cargo, tanker, windmill, platform and harbor structure.

TensorFlow [26] is a widely used machine learning library created by Google that aims to facilitate distributed machine learning using a graph structure. It is a marked improvement from the previous system Google had built, called DistBelief. It utilizes a unified structure between the data that flows through the graph and the computation graph operations themselves; both represented as nodes in a graph. This unified structure allows a higher level of abstraction than is normally achievable with machine learning libraries.

Keras [27] is an open-source framework that extends TensorFlow to provide ready-made machine learning components and layers than can then be easily combined to create complex workflows. Keras also supports alternative back-end systems like CNTK (Microsoft Cognitive Toolkit) or Theano, which are available as libraries providing similar primitives to TensorFlow.

IV. Our Approach

Figure 1 describes our workflow, that consists of several processing steps. These steps can be categorized as follows: (i) data correlation, (ii) processing and balancing, and (iii) training.

We used the Python package Sentinelsat[2] to directly download the Sentinel-1 SAR satellite images from the Copernicus database. For the data processing tasks that are described below, we used the Geospatial Data Abstraction Layer (GDAL) python library, which is commonly used for geospatial operations. The process begins with re-projecting the satellite image to a standard projection (EPSG:4326). Afterwards, the land masking of the satellite image is performed in order to remove the land areas depicted in the image, which are out of interest and might affect the accuracy of our framework. As an input to the process, a polygonal shapefile that includes the area of the satellite image is required so as to mask out the land. This shapefile should include a polygonal area which represents all of the land of the area involved. Then, the shapefile is projected to the same projection system as the satellite image, clipped to the corresponding area, and then used to mask the land.

The AIS positions are then processed via linear spatiotemporal interpolation, so that the ships are in the approximate positions they were when the satellite image was taken, and subsequently are transformed into geometries. We use the AIS data to automatically annotate the images with respect to the presence or absence of a ship in a fragment of the image (a tile). We thereby show how a carefully designed data fusion process can be used resolve the problem of data labeling - oftentimes the bottleneck in many Machine Learning applications.

The next step is the correlation task, which is performed by splitting the satellite image into tiles. Each tile retains metadata about its corner coordinates, and thus after the tile generation, the tiles area is transformed into a geometry, and the intersection of the resulting geometry is calculated for each AIS position. That is to say, each AIS point geometry (which can optionally have a larger radius than the point itself, to account for irregularities and azimuthal ambiguities), is checked against the geometry that represents the area of each tile, so as to check whether the ship – according to AIS – is contained within the tile. The tiles are then marked accordingly for consumption by the neural network. In this

[2]https://github.com/sentinelsat/sentinelsat

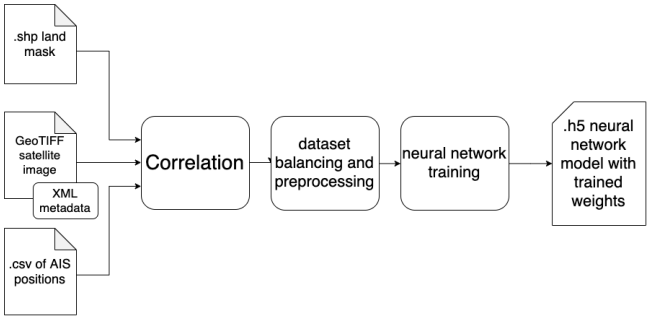

Fig. 1: Process workflow for correlation and training

way, AIS is considered the ground truth for the training of the neural network. This action is performed with every resulting tile. One of the parameters as input are the dimensions of the resulting tiles, which can be variable. Positions that are outside the bounds of the image post-reprojection and post-interpolation are discarded, and a distinction is made between truly-empty tiles (which have an average brightness of 0) and tiles that simply do not contain vessels (but contain background noise, waves, etc.). The former type of tile exists as due to the re-projection sections of the resulting image are blank. Tiles of the former type are discarded.

Ultimately the task is modeled as a classic binary classification problem on satellite images (we use Sentinel 1 SAR images in this paper). The image classification task itself is handled by the CNN algorithm (see Section III). The result of this process is a set of ready-marked tiles for consumption by the convolutional neural network. In essence, the goal is for the neural network to be able to recognize the presence of a ship within a certain section of the satellite image. Specifically, it is a binary classification task of "ship(s) present/ship(s) not present". This is why in Section V, metrics that are only relevant to binary classification ML problems are given, such as Area under ROC Curve, Precision, Recall, etc. A convolutional neural network is used, instead of any other type, due to its inherent feature learning ability, i.e. its ability to capture relevant features from an image on different levels, similar to the human brain.

V. Experimental evaluation

To evaluate the methodology several satellite images from the EU Copernicus project were used, specifically from the Sentinel-1 satellite. Five images were used for the training, each containing approximately 20-30 ships, and for the testing a single hand-correlated tileset containing 73 representative samples of ships was used. Sentinel 1 is equipped with a SAR sensor, so the output images have the advantage of being unaffected by weather conditions due to the radar nature of the sensor. The images were taken primarily from two areas, south of the Iberian peninsula and the Aegean, due to both the fact that it has excellent terrestrial AIS coverage (to ensure that no ships are missing from the AIS data) and to the large amount of ship traffic that passes through both these areas, ensuring that the ships-per-image ratio was fairly high so as to extract as much value training-wise from each image as possible. The AIS data for each area was sourced from MarineTraffic.[3]

The testing of the neural network was done on an image hand-correlated via QGIS taken from the Aegean sea, and as such it is certain that the labelling was done without any correlation errors. To perform this correlation by hand, the image was loaded into the QGIS software, and the AIS positions were overlaid on top of the image. The image was then split into tiles, and each tile that corresponded to an AIS position was marked at first as a ship tile. At this point, the process is the same as that which is done automatically through the methodology. However, the correlation by hand involved the further step of correcting the automatic process when the AIS positions did not correspond, due to azimuthal ambiguities or imprecise spatiotemporal interpolation, to the underlying position of the ship in the image, ensuring 100% accuracy in the correlation, something the automatic process does not guarantee.

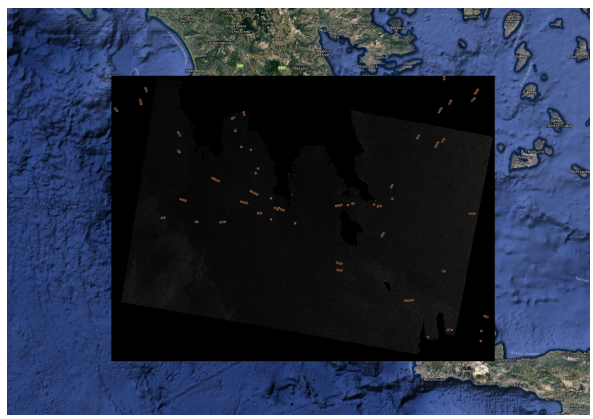

Fig. 2: Example SAR image, taken from around the Peloponnese, with AIS positions overlayed

VI. Neural Network Architecture

In the context of this work, as explained earlier, we used a convolutional neural network built using Keras. It is composed of 3 convolutional layers and 2 fully-connected layers, as shown in Figure 3. Implicitly between each layer of the figure exists a ReLU activation layer. As an optimizer, Nadam was used (Adam optimizer with enhanced Nesterov momentum– one of the baked-in optimizers Keras offers), with a standard learning rate and other parameters. Dropout layers were used as well. This structure was experimentally found to bring the best results. As supervisory metrics, accuracy, binary cross-entropy loss, precision, recall, and F1 score were used. Early stopping was utilized to optimize the network's accuracy. In addition, due to the relatively small size of the dataset, transformations of horizontal and vertical flipping were used on each tile so as to increase the training ability of the network. Accuracy and loss metrics were recorded and graphed, as well as ROC curves plotted.

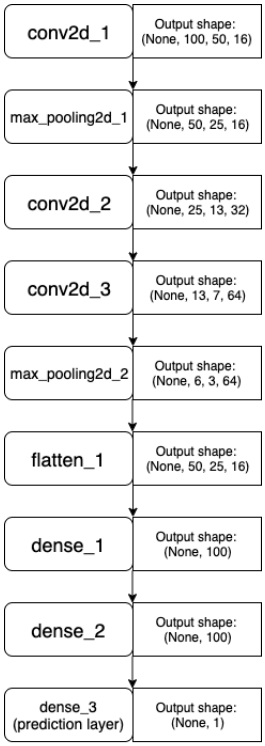

Fig. 3: Architecture of CNN

Threshold	0.30
Area under ROC score	0.946
Accuracy (%)	0.884
Accuracy (number of correct identifications from test set)	114/129
Precision	0.892
Recall	0.904
F1 Score	0.898

TABLE I: Results from best model

VII. Results and Discussion

The results are fairly promising. Compared to the model loss graph shown in Figure 5 and the model accuracy graph shown in Figure 4, the actual loss and accuracy are much higher. This is due to the fact that these graphs were generated based on the automatically correlated data, while the testing was done on the manually-correlated image as mentioned in Section V, which has a 100% guarantee of correct correlation.

In addition, due to the somewhat-limited nature of the training dataset, in both Figure 4 and Figure 5, some over-fitting can be observed. Despite attempted strategies to mitigate this factor, such as Dropout layers and Batch Normalization, it was unfortunately not mitigated fully. Regardless, despite the over-fitting, the results on the hand-correlated test dataset remained excellent.

The threshold value, as shown in the table above, is the value above which the prediction of the neural network is considered to belong to the predicted class. For example, if the neural network gave an output value of 0.7, if the threshold were 0.5 it would be considered a positive result, but if the threshold were 0.8 it would not. In this way, by tuning the threshold value, it is possible to adjust whether higher recall or higher precision is given more weight, which is generally selected based on the nature of the problem at hand.

[3]http://www.marinetraffic.com

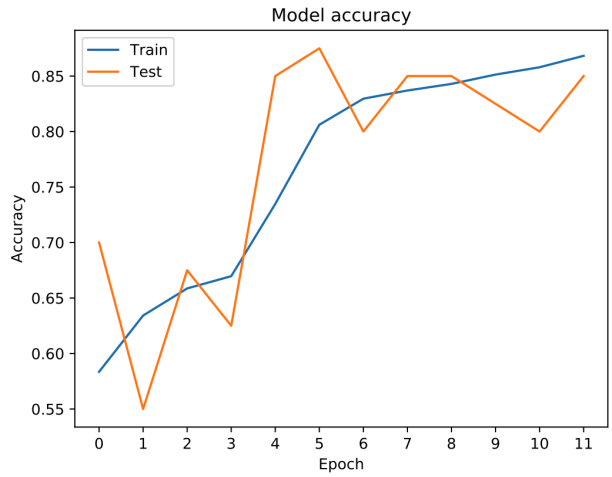

Fig. 4: Accuracy progression during training

When tuning the neural network threshold in this case, higher recall (i.e. lower threshold) was preferred to higher precision. This is because in a real-world scenario, it would be more damaging to miss a ship (i.e., identify that no ship is in a region where one is, potentially allowing illegal activity to go unnoticed) than to classify as a ship something that is not (i.e., identify that a ship is in a region where there is nothing there). This also explains the discrepancy that seems to exist between the Area under ROC score (from Figure 6) and the accuracy. A higher accuracy could be achieved, as well as a higher (near perfect or perfect) precision, if a sacrifice was made with regards to the recall.

This led to the result of higher testing accuracy vs. the training and validation datasets, as the faults in the automatic correlation leads to noise in these datasets. This constitutes the primary challenge for this problem, as both the azimuthal ambiguities inherent in the SAR sensor and the occasional incorrect spatiotemporal interpolation lead to occasional incorrect correlations between AIS positions and the underlying image. The spatiotemporal interpolation is linear, and as such if the ship greatly changed heading or position this could affect whether the correlation is correct or not. It should be noted that the approach and methodology is fully generalizable to other types of satellite data, e.g. non-radar based/optical types, in which case this noise would not be present and the accuracy would likely be higher.

VIII. Conclusions and Future Work

In this paper we describe a fully automatic framework for deriving information about the existence of vessels in Sentinel 1 SAR imagery. By fusing satellite and AIS data, we semantically annotated the images about the existence of vessels and we created automatic training and test datasets which were provided as input to a neural network. Thus, our neural network is able to detect vessels in satellite images

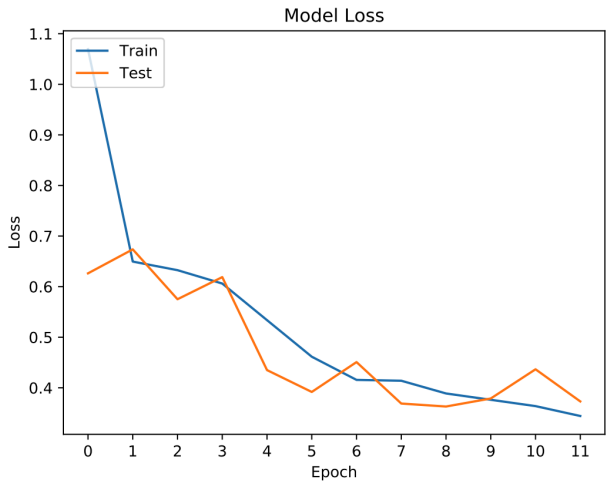

Fig. 5: Loss progression during training

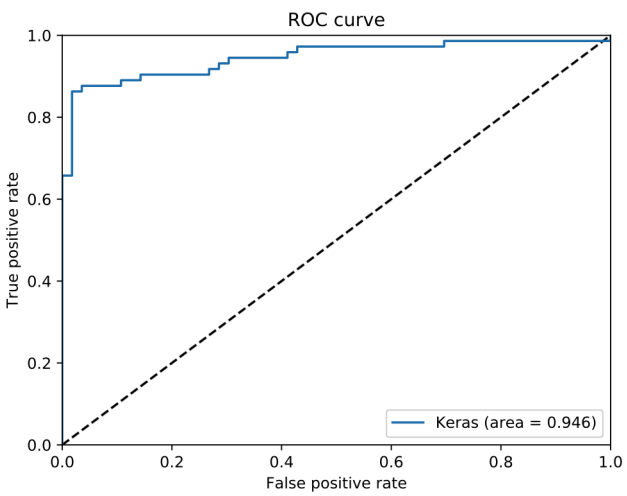

Fig. 6: ROC curve for best model

providing 89% accuracy and an Area under ROC score of 0.94.

Regarding future work, the approach presented in this paper could be extended as follows: (i) adding more feature predictions, and (ii) more data sources. Other feature predictions could be performed using the same process, if that feature exists as part of the AIS dataset that is used for correlation. A likely candidate for this would be the ship type, as this information is available in the AIS data already. Alternatively, another candidate would be predicting the heading. However, using SAR imagery to make predictions about this kind of features is very challenging. The reason for that is that SAR images do not have the required resolution and quality so as to predict features other than simple presence; as shown in Figure 2, where it is visible that each ship is simply a dot, from which to predict anything like type or heading would be extremely difficult.

The code and methodology that this paper describes is not specialized to SAR imagery; the same code and methodology could function on other types of satellite imagery, for example optical imagery or imagery from drones. SAR comes with issues that affect the quality of correlation, such as azimuthial ambiguities and radar backscatter, that would not be present from sources such as optical satellite imagery. However, optical satellite imagery would have other issues, such as clouds and weather phenomena, that would not affect the correlation itself per-se (the ships would be present underneath in the right place, e.g. under the clouds) but would affect the neural network's learning (given that the ships would be obscured), and as such the final accuracy would still be affected if the images used were not vetted.

There are two further limitations of this work: the update frequencies of the various data sources (satellite and AIS), and the nature of the AIS training data. On the point of the satellite update frequency, the Sentinel-1 satellites visit and photograph the same place on Earth about once every 24H. This means that more granular examination in terms of smaller time slices is not possible, as the update frequency is this low, and that the only potential use would be to examine events post-facto, and not as they are occurring. However, in addition to this, AIS has a comparatively much better, but still not constant update frequency. Therefore, a factor that affects the quality of the correlation and by extension interpolation is the AIS coverage in the area, and the frequency of the signals that the ships are sending (which is something variable and determined by each ship individually). The larger the time delta from the AIS signal and the timestamp of the satellite photograph, the more inaccurate the interpolation will be, due to the unpredictability of ship movement. The second limitation has to do with the nature of AIS data: as noted earlier, AIS does not capture all types of ship data. Smaller ships that are not required by law to carry an AIS transceiver are not picked up, as well as ships that have AIS but then shut it off. Therefore, hypothetically, the system is learning to ignore these types of ships, as they are not visible in the data that it is training on (i.e. AIS). The potential way to mitigate this would be to use areas that are likely low in these types of events. As well, if the frequency of these events is low enough, the network will still be able to see through the noise and learn the correct solution.

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